

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 02

Sequential Circuit



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Recap of Previous Lecture



S-R latch

S-R ff





Topics to be Covered

J-K ff

Concept of triggering

- Problem with S-R flip-flop :

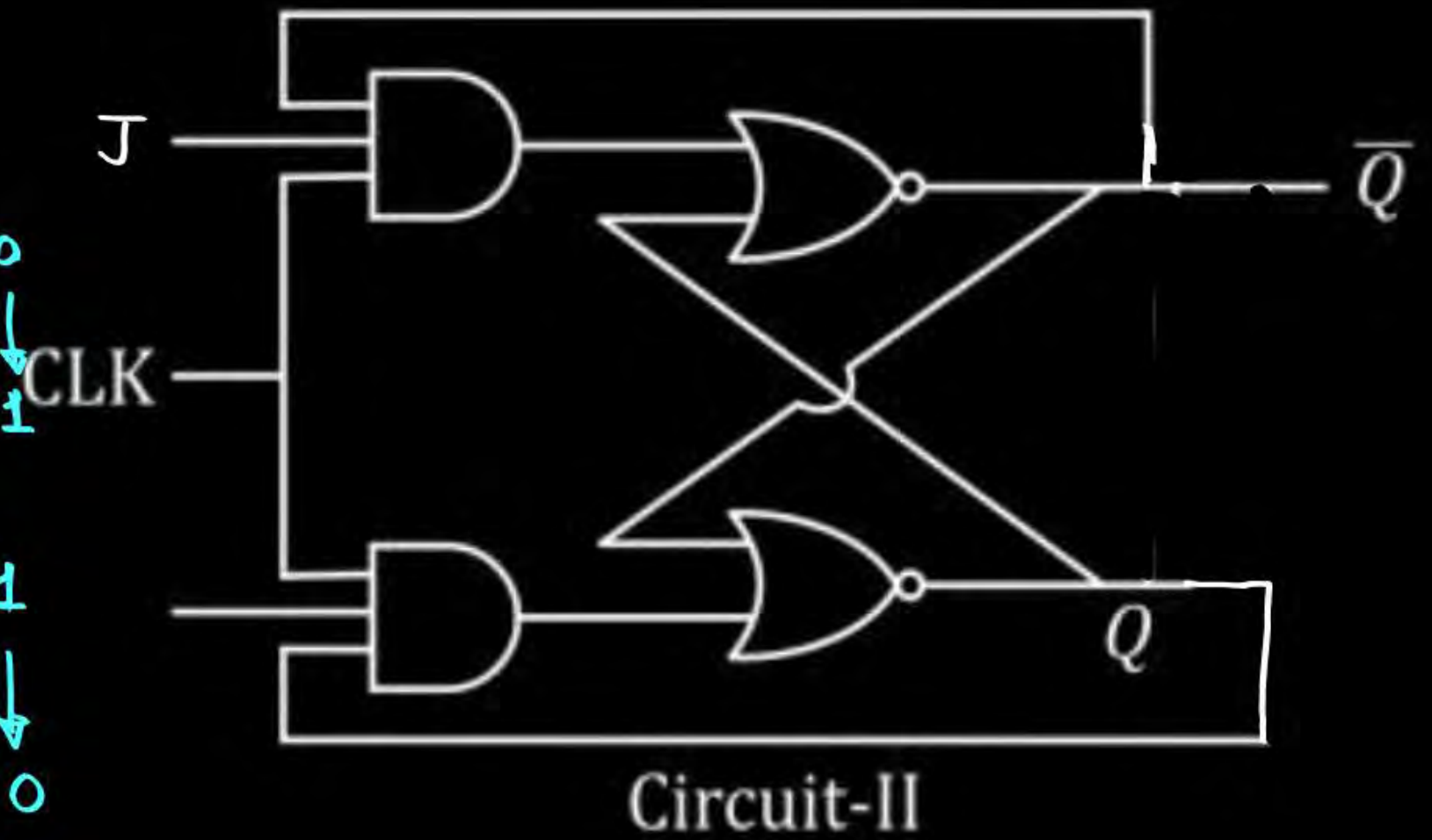
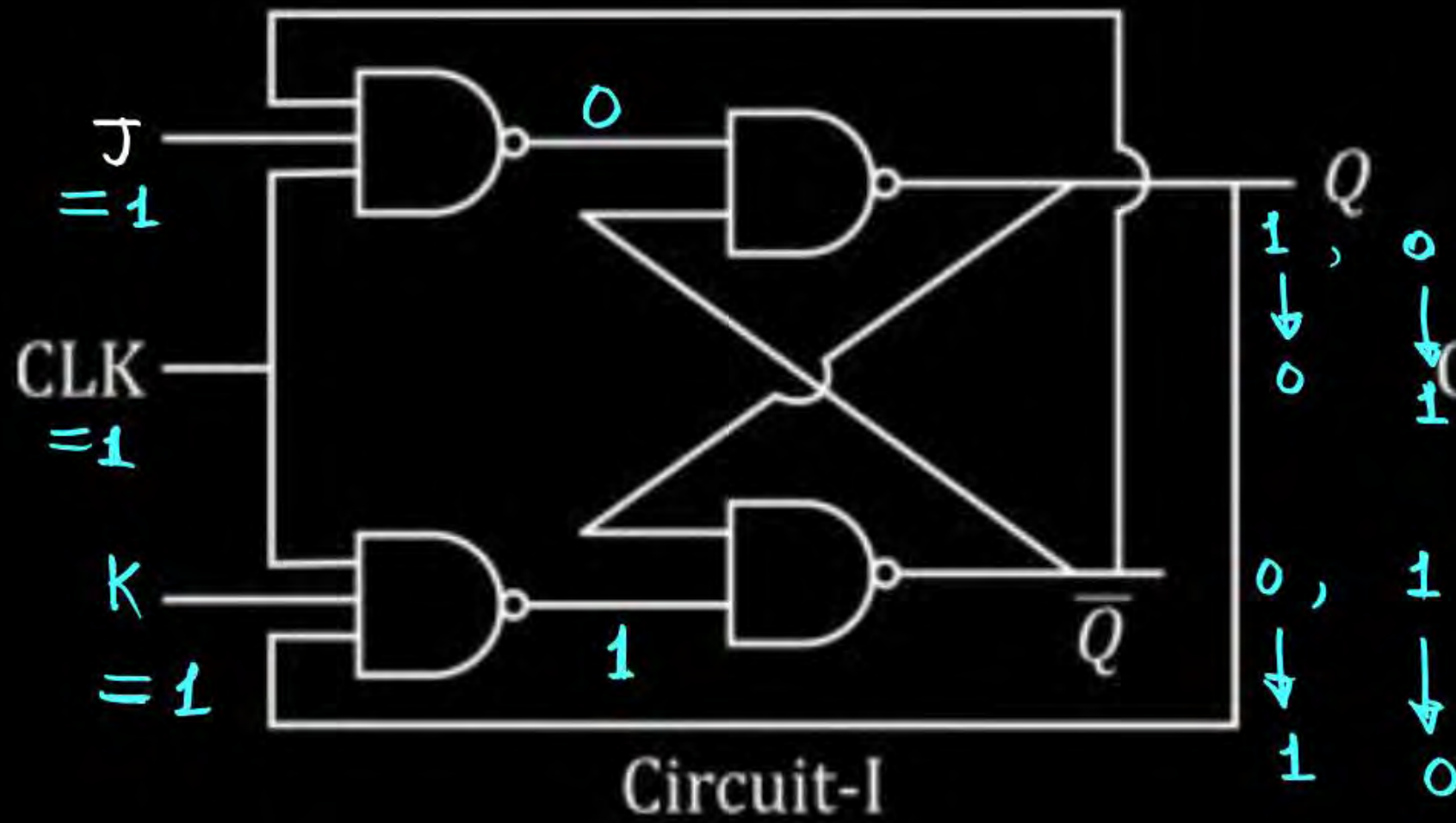
$S=1$ & $R=1$, generates invalid O/P [Q & $\bar{Q}$ become same] and this creates problem in designing of sequential ckt.

↓
To remove this problem we design J-K ff.

[J-K Flip-Flop]



- Designed to avoid invalid state of S-R Flip-Flop.
- Circuit:



- Working :

Clock	J	K	$Q(n+1)$
Inactive	X	X	$Q(n)$
Active	0	0	$Q(n) \longrightarrow$ hold state
Active	0	1	0
Active	1	0	1
Active	1	1	$\overline{Q(n)} \longrightarrow$ toggle state

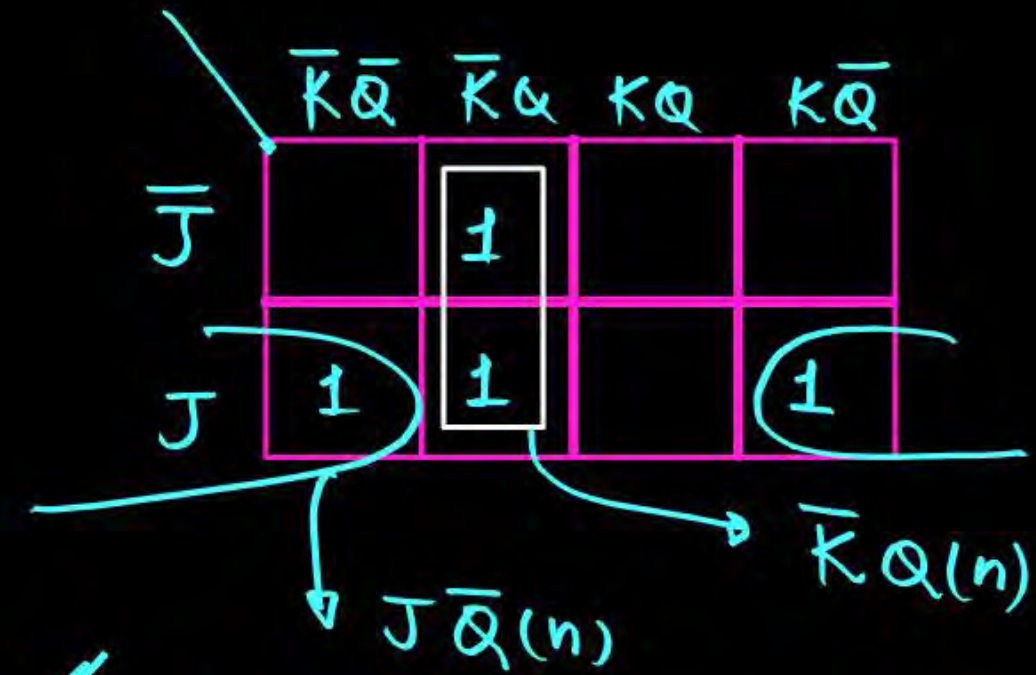
Characteristic Table :



$$Q(n+1) [J, K, Q(n)] = \Sigma(1, 4, 5, 6)$$

Clock Active

J	K	$Q(n)$	$Q(n+1)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



$$Q(n+1) = J\bar{Q}(n) + \bar{K}Q(n) \rightarrow \text{always valid when clock is active.}$$

Excitation Table



$Q(n)$	$Q(n+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

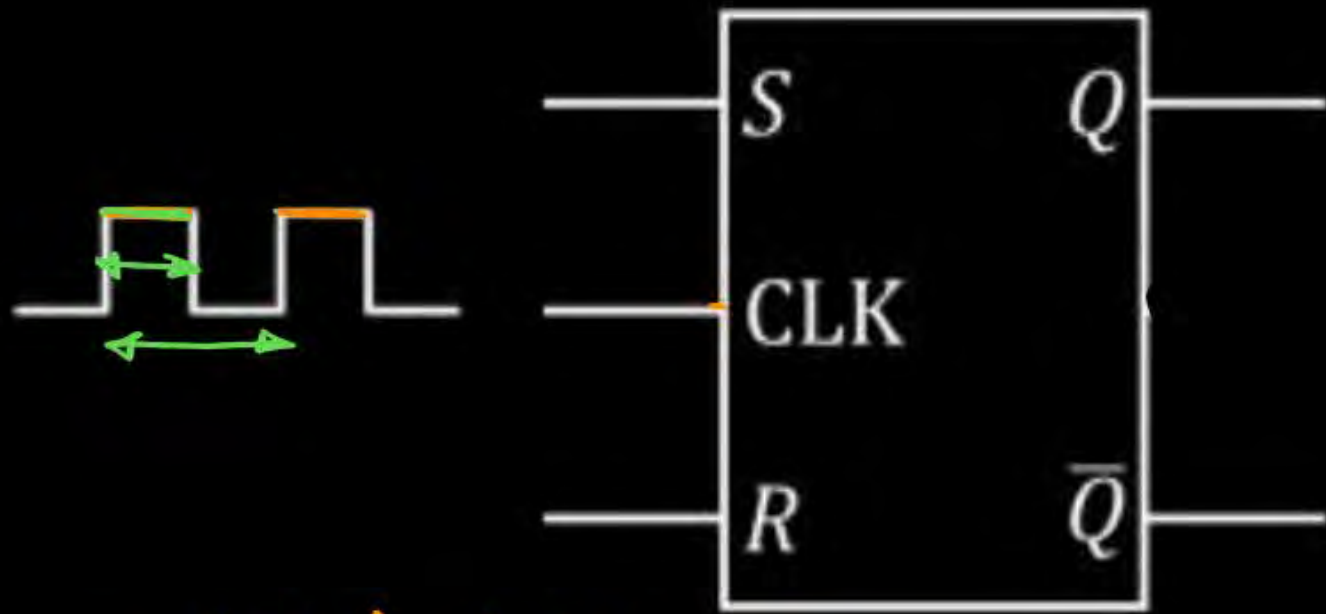


0	$\xrightarrow{J=0, K=0}$	0
0	$\xrightarrow{J=1, K=0}$ $\xrightarrow{J=1, K=1}$	1
1	$\xrightarrow{J=0, K=1}$ $\xrightarrow{J=1, K=1}$	0
1	$\xrightarrow{J=0, K=0}$ $\xrightarrow{J=1, K=0}$	1

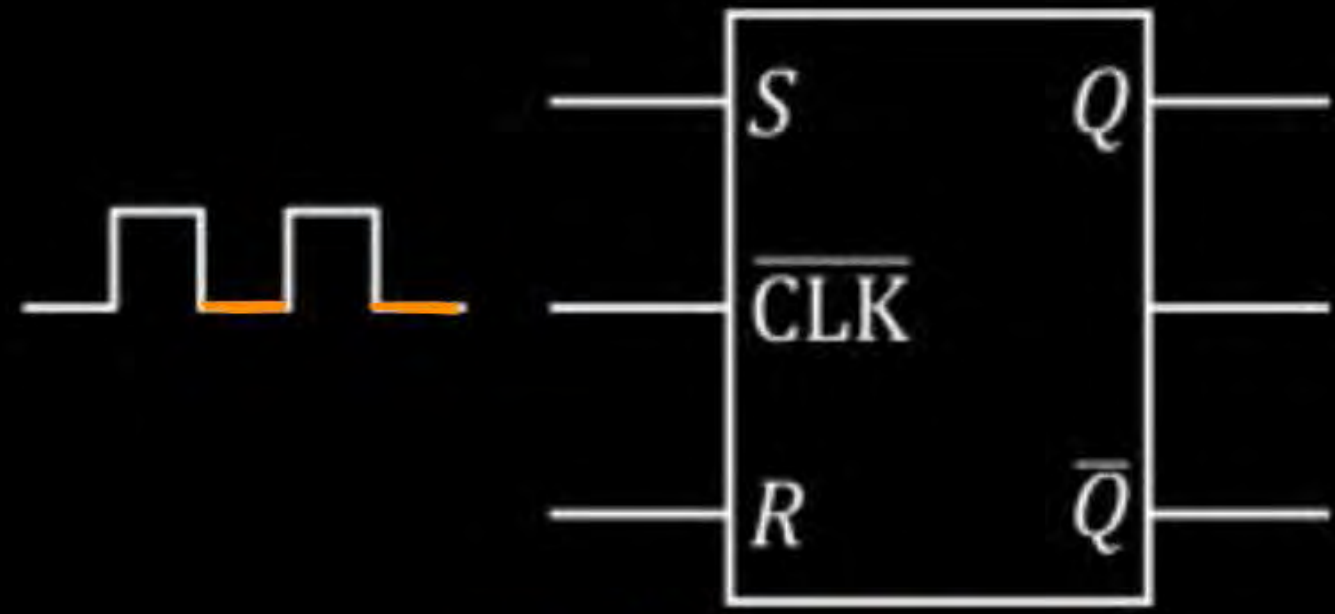
[Level Trigger S-R Flip-Flop]



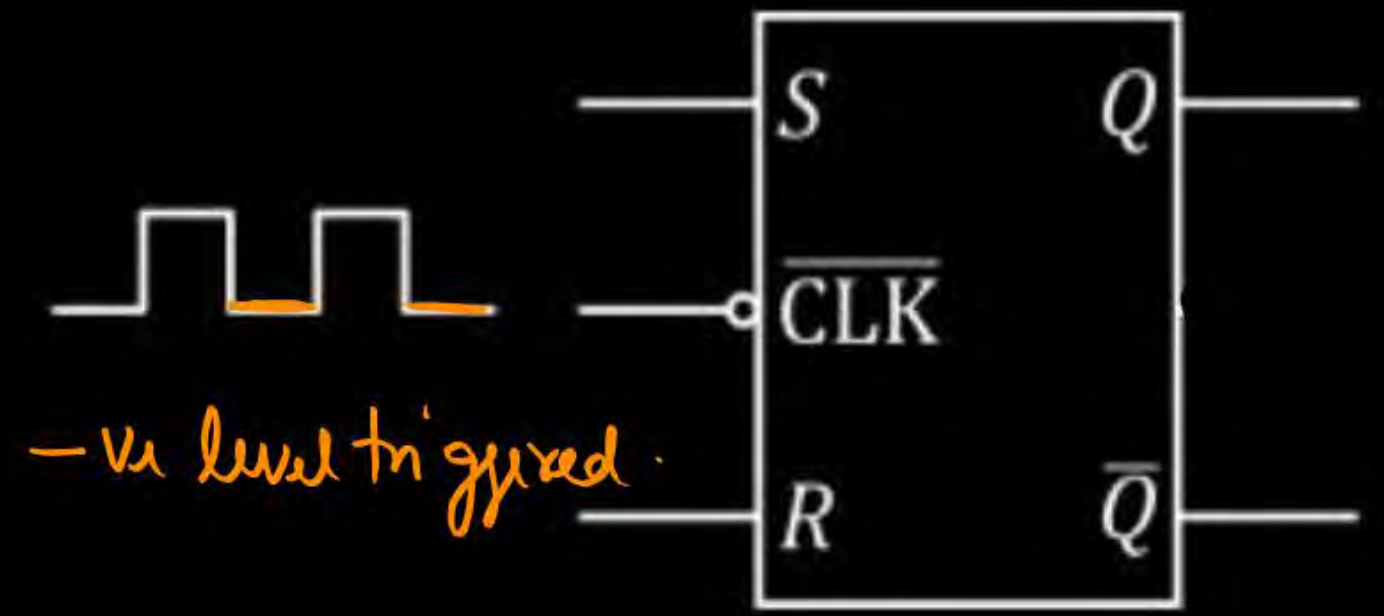
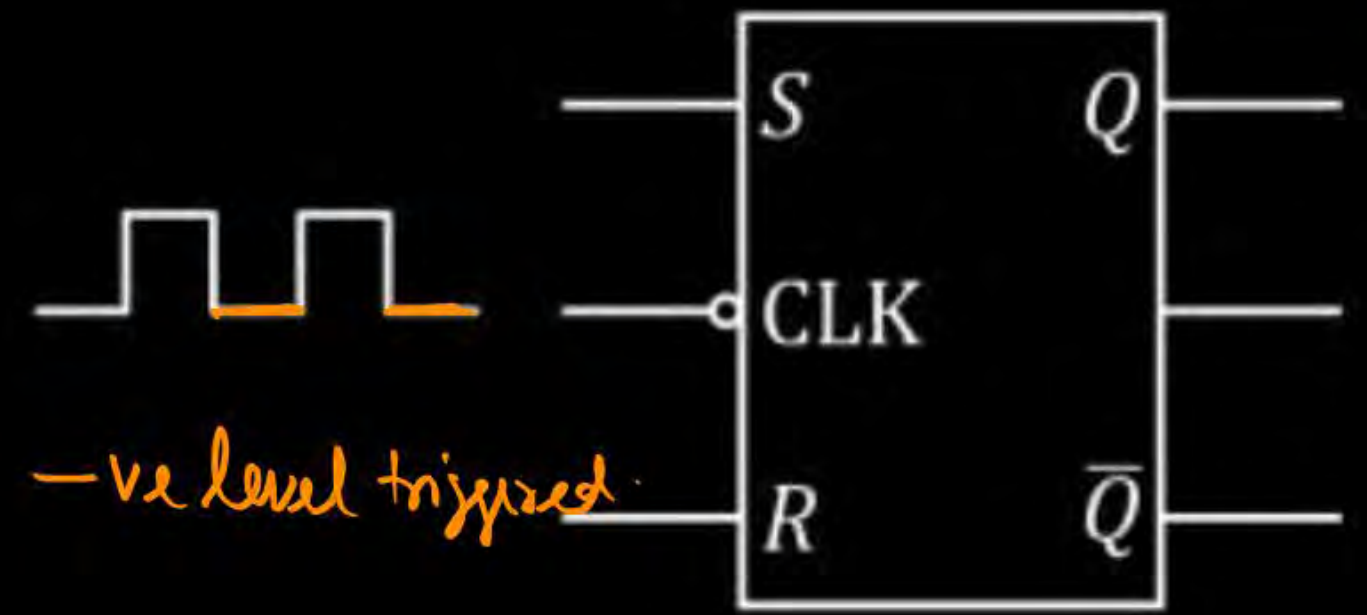
Different representation of level triggered FF's



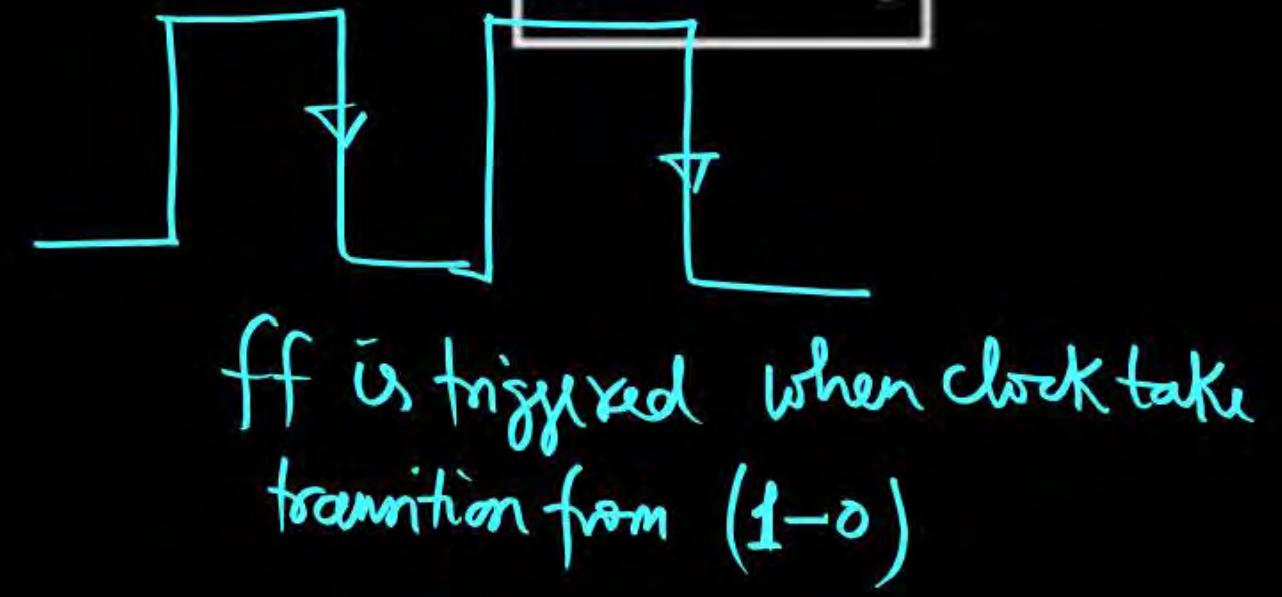
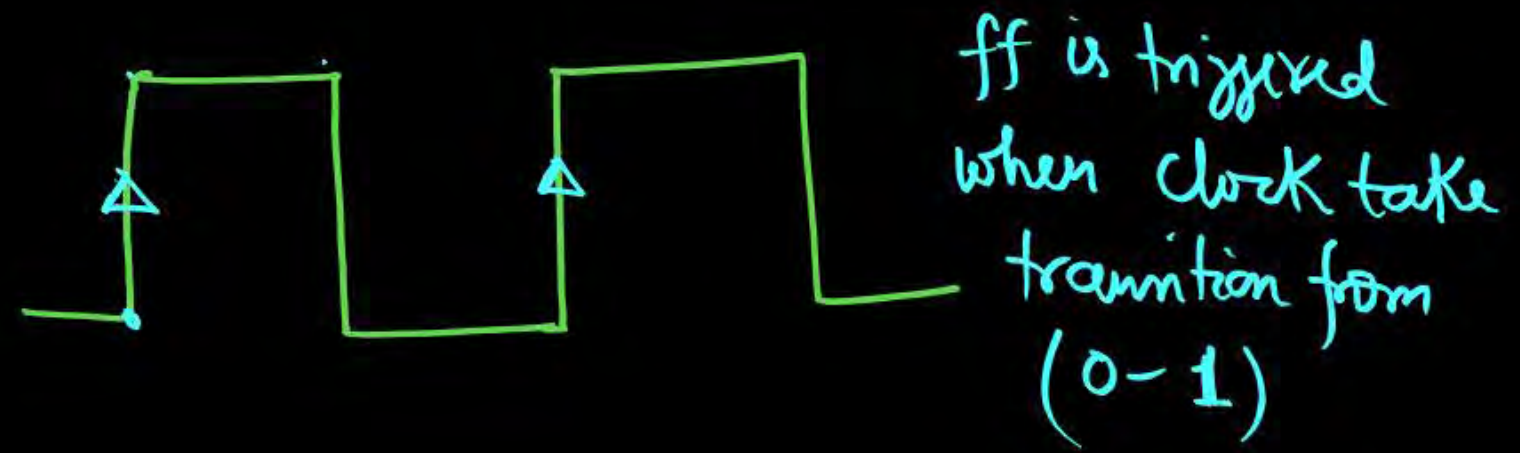
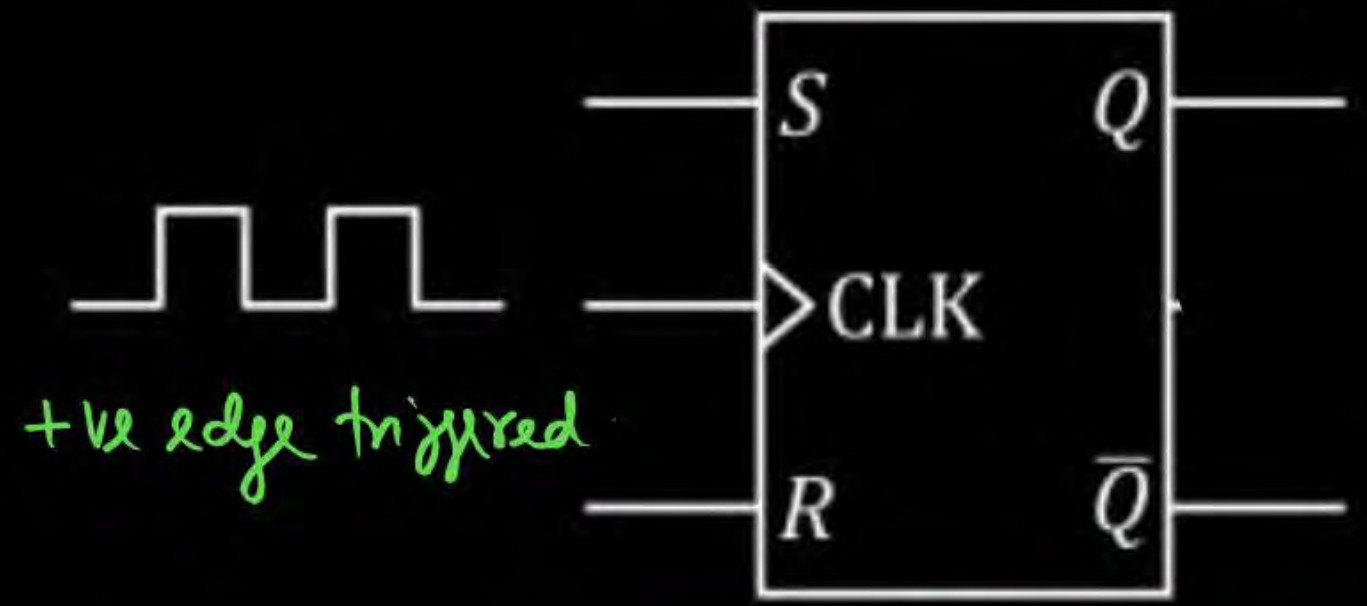
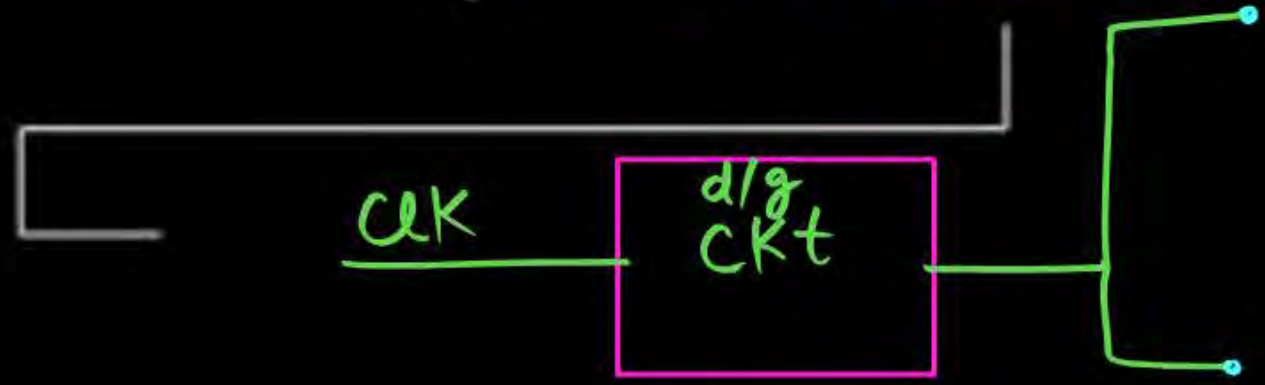
+ve level triggered

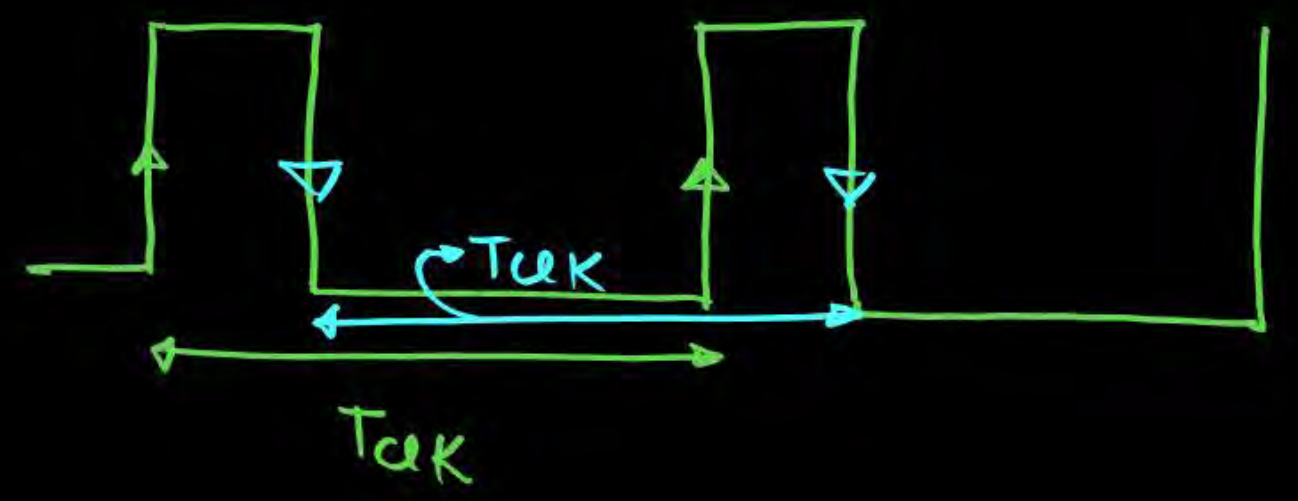
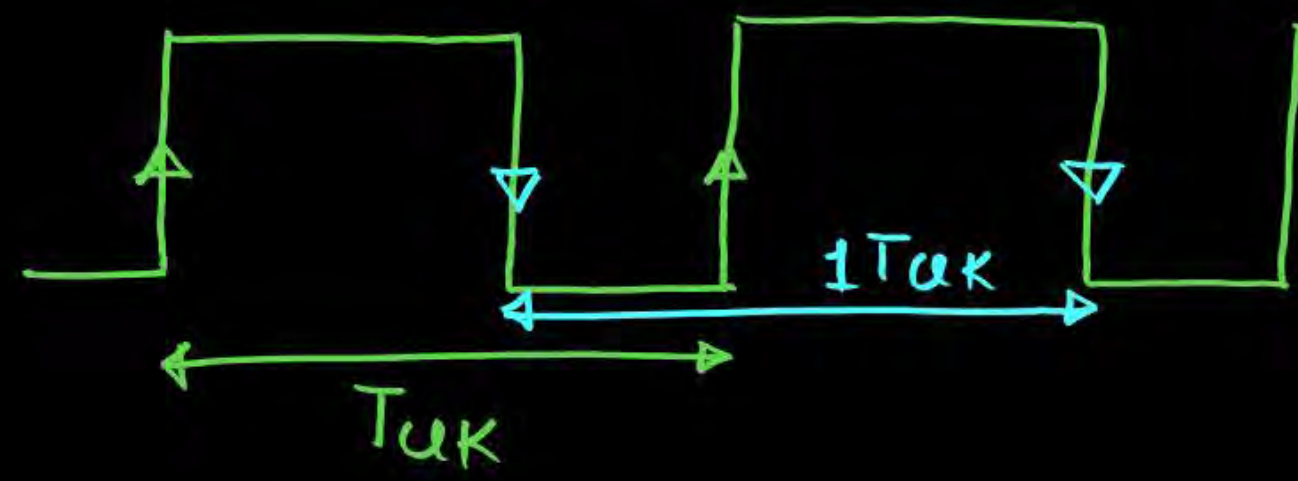
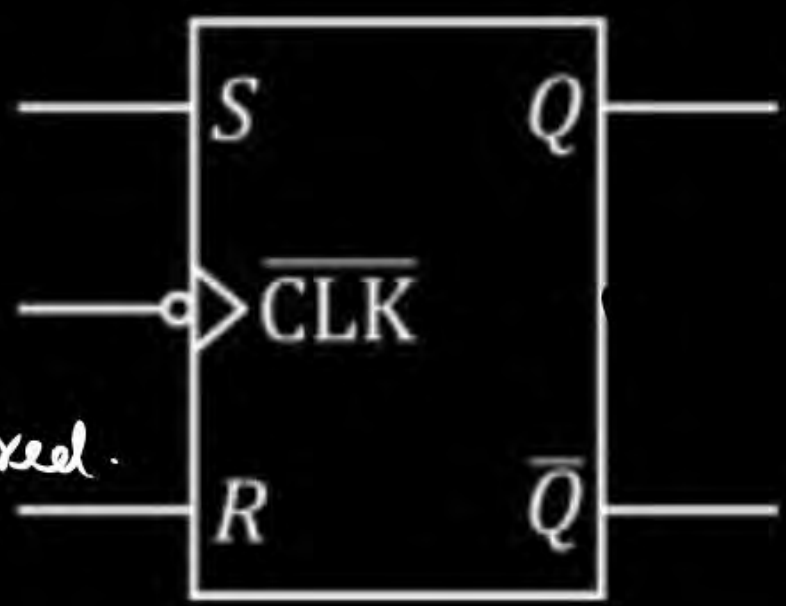
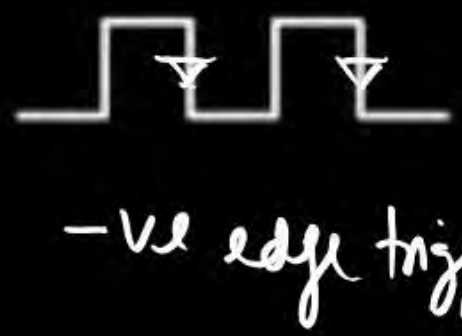
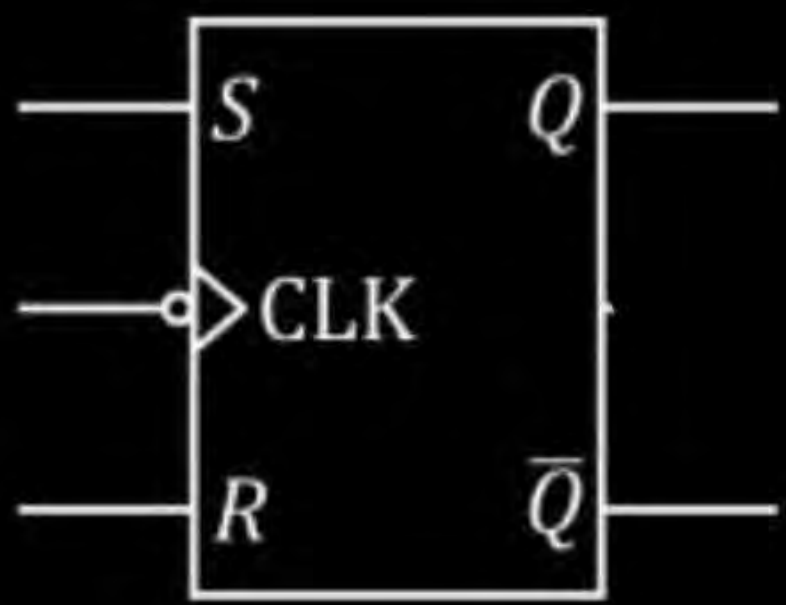
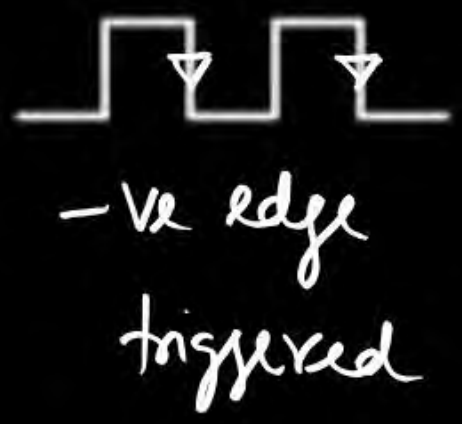


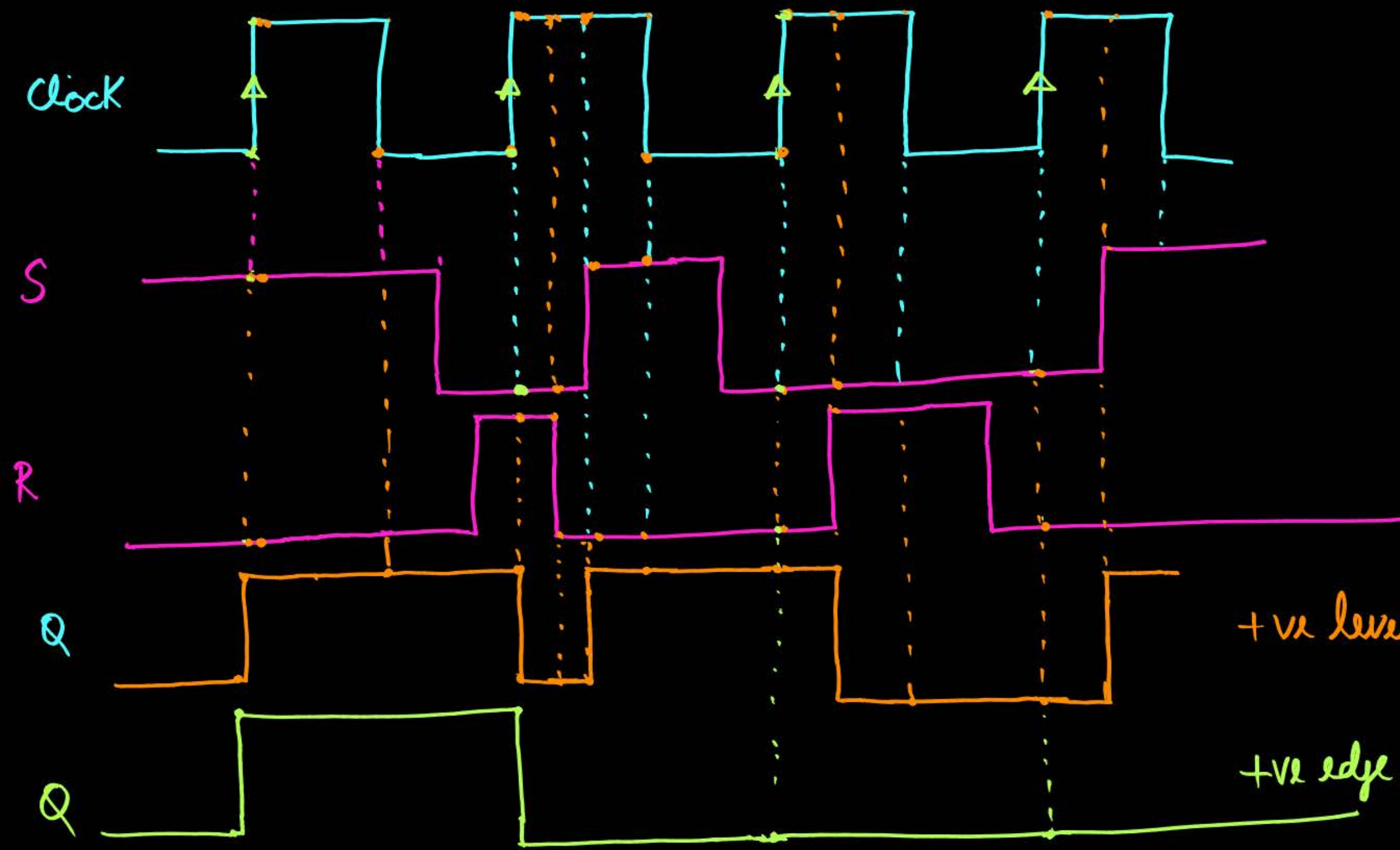
-ve level triggered



[Edge Triggered FF]







H.W.

→ -ve level S-R ff
→ -ve edge triggered ff

+ve level triggered S-R ff

+ve edge triggered S-R ff

- IMP points about level triggered and edge triggered :

- In level triggered ff O/P may change more than once in a single clock cycle.
- While in case of edge triggered ff O/P changes only once in a single clock cycle.

[Race Around Condition]



- Lets understand what is Race-around condition and when it can happen :

level triggered J-K ff

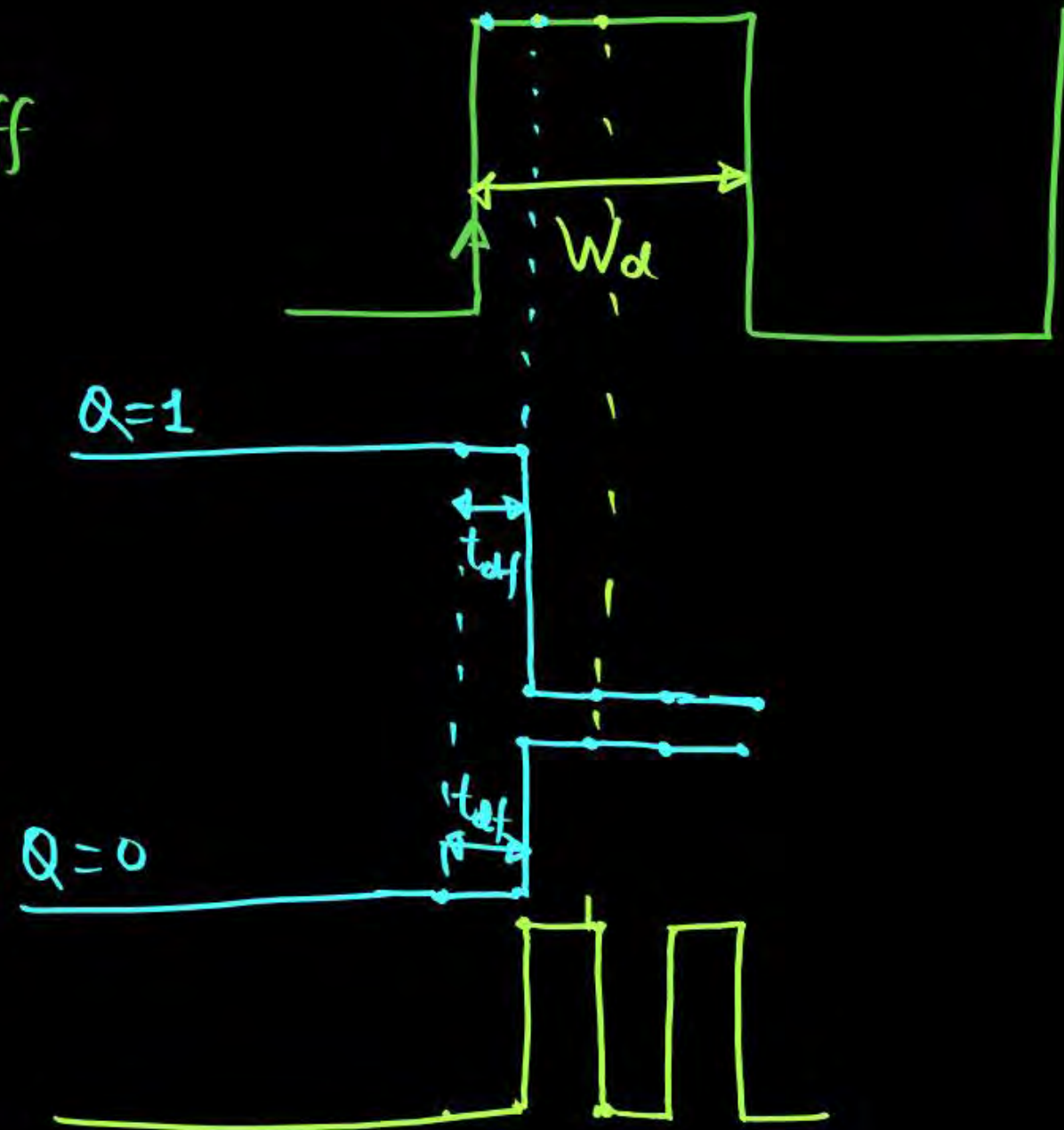
$J=0, K=1$

$Q=1$

$J=1, K=0$

$Q=0$

$J=1, K=1$



$$W_d \gg t_{df}$$



Topic : 2 Min Summary

- S-R ff
- J-K ff
- Concept of triggering.

Thank you

GW
Soldiers !

